HOLLOWAY INTRALOGISTICS CONSULTING LIMITED

Independent warehouse-automation & intralogistics performance experts

IN THE HIGH COURT OF JUSTICE — COMMERCIAL COURT (KBD) — Claim No. CL-2025-000287

Penrose Robotics Limited v Caldervale Distribution Limited

EXPERT REPORT OF DR ALAN HOLLOWAY

On the cause of the throughput and availability shortfall during the Acceptance Period

1. Instructions and expertise

1.1 I am Dr Alan Holloway, Director of Holloway Intralogistics Consulting Limited. I hold a PhD in control systems and have 22 years' experience in the design and performance assessment of automated fulfilment systems, including goods-to-person robotic deployments. I am instructed by Whitfield Carter LLP on behalf of the Claimant. I understand my overriding duty is to the Court and I have complied with CPR Part 35 and the associated Practice Direction.

2. The questions I address

2.1 I am asked: (a) whether the Penrose System was, as designed and supplied, capable of achieving 650 units/hour/zone and 98% availability; and (b) what caused the measured shortfall during the Acceptance Period of 7 October to 1 November 2024.

3. Material considered

3.1 I have reviewed the OrchestraOS acceptance-test logs (per-zone, per-hour throughput and availability telemetry), the SKU master files of 18 and 26 September 2024, the host-environment monitoring data, and Caldervale's DC throughput records. Where I rely on raw telemetry I have reproduced the underlying figures in Appendix A.

4. Capability of the System

4.1 Analysing the 1,920 zone-hours of logged data, I find that in periods where (i) the SKU records used were dimensionally accurate and (ii) host latency was below 8ms and (iii) the order profile was within the agreed test envelope, mean throughput was 671 units/hour/zone (n=214 zone-hours), comfortably above the 650 threshold, and availability in those periods was 98.4%. This establishes that the System as supplied was inherently capable of the contractual performance.

5. Cause of the shortfall

5.1 The blended Acceptance-Period averages relied on by Caldervale (521 units/hour/zone; 93.8% availability) are depressed by three identifiable factors. Using a regression decomposition

(Appendix B), I attribute the throughput shortfall of 129 units/hour/zone approximately as follows: SKU-master inaccuracy ≈ 58 u/h (45%); host-latency throttling (≈31ms vs ≤5ms specified) ≈ 47 u/h (36%); and the out-of-profile promotional surge from 21 October 2024 ≈ 24 u/h (19%). Each of these factors is, in engineering terms, attributable to the host site rather than to the robotic system or OrchestraOS.

5.2 Inaccurate SKU dimensions cause the orchestration engine to mis-estimate tote fill and to generate sub-optimal robot routes, with a direct and quantifiable throughput penalty. Host latency of ~31ms lengthens the order-release cycle and reduces effective fleet utilisation. The promotional surge introduced a high proportion of single-line micro-orders that the agreed test profile expressly excluded.

6. Availability

6.1 The availability shortfall (98% target vs 93.8% measured) is largely explained by charge-cycle disruption arising from the same routing inefficiencies, compounded by two host-side network outages on 17 and 24 October 2024 recorded in Caldervale's own infrastructure logs. I find no evidence of a systemic fleet reliability defect.

7. Conclusions

7.1 In my opinion, on the balance of probabilities, the Penrose System was capable of meeting the Acceptance Criteria, and the measured shortfall during the Acceptance Period was caused by Caldervale's SKU-master inaccuracy, its non-compliant host environment and the out-of-profile promotional surge, rather than by any defect in the System as supplied. Had those three factors been absent, I would expect the System to have achieved the contractual throughput and availability.

Statement of truth

I confirm that I have made clear which facts and matters referred to in this report are within my own knowledge and which are not. I confirm that the opinions I have expressed represent my true and complete professional opinion. I understand my duty to the Court and have complied with it.

Dr Alan Holloway — Holloway Intralogistics Consulting Limited — 26 September 2025